



Lecture 4: Storage and flexibility

Energy Economics of the Green Transition

Rasmus K. S. Berg

Monday 28 September 2026

Department of Economics, University of Copenhagen

1. From costs to value: what generation costs, and what it is worth
2. Four ways to balance a system
3. Storage technologies
4. A workshop on Zerrahn et al. (2018): how much storage does a renewable system need?

From costs to value

Four kinds of cost

- **Investment**, EUR per kW: building the plant. Paid up front, proportional to its size, and *sunk*.
- **Fixed operation and maintenance**, EUR per kW and year: keeping it available, whether or not it runs.
- **Variable**, EUR per MWh: fuel, carbon and wear,

$$c = \frac{p^F}{\eta} + o \quad \text{and} \quad e = \frac{\varphi}{\eta} \text{ t CO}_2 \text{ per MWh.}$$

The marginal cost: it sets the merit order.

- **Adjustment**: what it costs to *change* output. Cold starts, ramps, minimum load. A cost of a pattern, not a level.

Generation Technologies, §2. The first three are lecture 3's model; the fourth is today's.

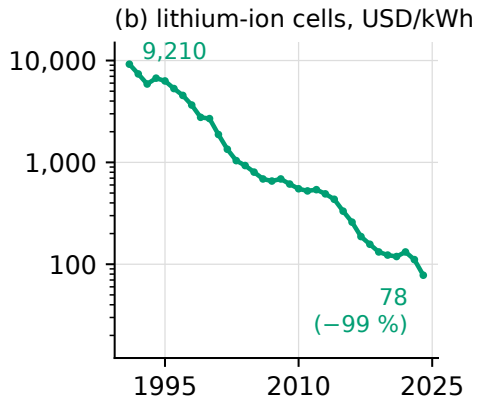
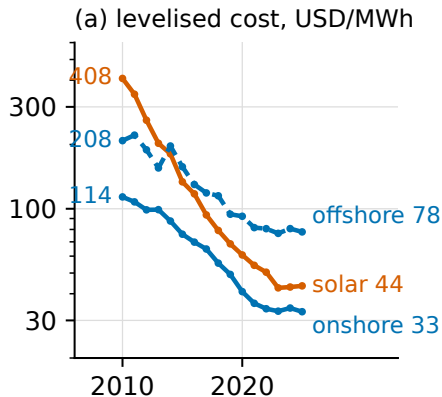
Technologies by cost structure

Technology	Investment	Running	Adjustment	Time to build
Solar PV, utility	Medium	Low	High	Low
Solar PV, rooftop	High	Low	High	Low
Onshore wind	Medium	Low	High	Medium
Offshore wind	High	Low	High	High
Hydro, reservoir	High	Low	Low	High
Hydro, run-of-river	High	Low	High	High
Nuclear	Very high	Low	High	Very high
Coal	High	Medium	High	High
Gas CCGT	Medium	Medium	Medium	Medium
Gas OCGT (peaker)	Low	High	Low	Low
Oil peaker	Low	Very high	Low	Low
Biomass CHP	High	Medium	High	Medium
Waste CHP	Very high	High	High	Medium

Generation Technologies, Table 1. Levels are relative to the other rows.

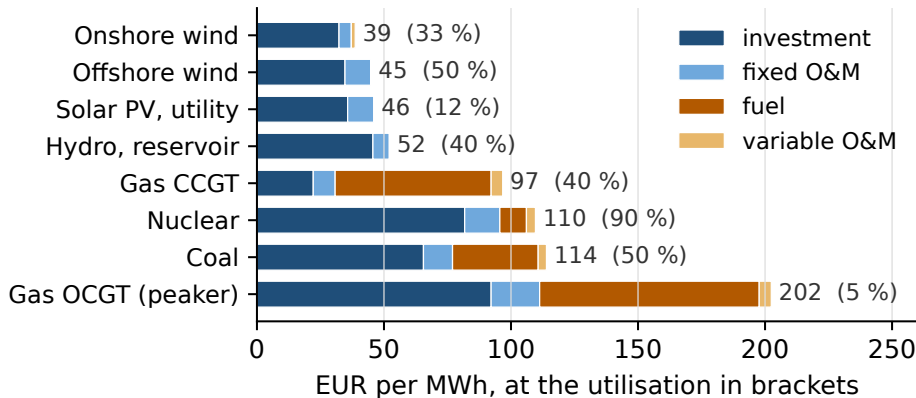
Which rows could balance a system with a lot of wind?

Learning curves



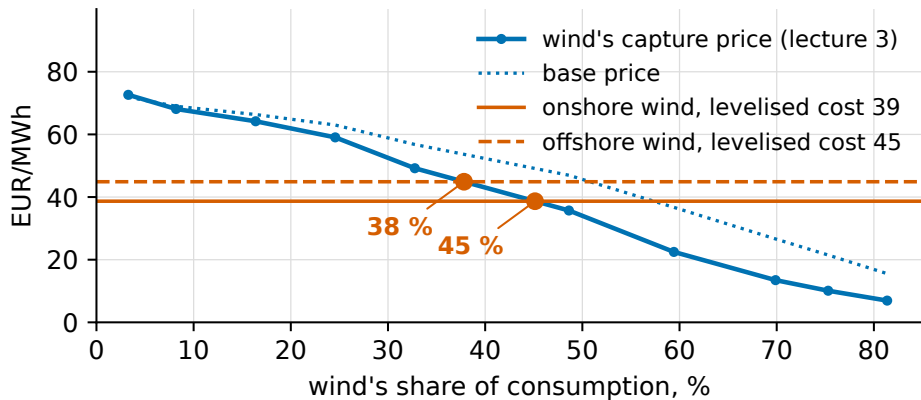
Our World in Data, CC BY 4.0: (a) IRENA, new projects worldwide, 2025 USD; (b) Way (2026), 2024 USD. Log scales.

By levelised cost, wind and solar have already won



Generation Technologies, Fig. 3.1: 2025 costs, the electricity rows.

Cannibalisation of revenue



Lecture 3's model, the fleet held fixed; levelised costs from *Generation Technologies*.

The hourly balance, and what can move

Every hour, in *Energy System Models'* notation:

$$\underbrace{\gamma_h \bar{q}_w}_{\text{wind and solar}} + \underbrace{q_{g,h}}_{\text{dispatchable plants}} + \underbrace{g_h^d - g_h^c}_{\text{storage}} = \underbrace{D_h}_{\text{demand}} + \underbrace{x_h}_{\text{net exports}}$$

- The first term moves with the weather. When it moves, some blue term must move with it, in the same hour.
- **Until recently:** demand was fixed, thermal plants did all the moving, and trade was at the margin.
- **As wind and solar replace thermal plants,** the terms that used to move are the ones being retired.

Four ways to balance

- **Flexible backup** ($q_{g,h}$): dispatchable plants raise their output when the wind drops, and lower it when the wind returns.
- **Flexible demand** (D_h): consumers move their use into the hours when power is plentiful.
- **Storage** ($g_h^d - g_h^c$): a surplus MWh is kept for a later hour, and some of it is lost on the way.
- **Trade** (x_h): a surplus is sold to the neighbours, and a shortfall is bought from them.

Today: storage, and its partner **curtailment**: throwing part of a surplus away.

A still, cold fortnight in January. Which option carries the system?

1. a large fleet of four-hour batteries
2. flexible demand: heat pumps and electric cars shift their use
3. backup plants and imports
4. curtailment

Storage technologies

A storage is two capacities

- **Power**, MW: how fast it charges and discharges. For a battery, the inverter.
- **Energy**, MWh: how much it holds. For a battery, the cells.
- **Duration** = MWh / MW: a 4-hour battery empties in four hours at full power.
- **Round-trip efficiency**: the share of a MWh put in that comes back out.

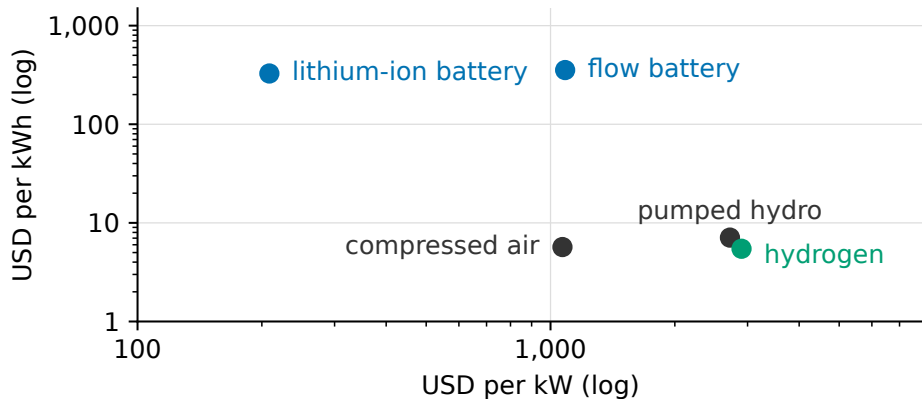
Energy System Models' battery

	2025	2050
inverter, EUR/kW	147	64
cells, EUR/kWh	128	80
4-hour system, EUR/kWh	165	96
round trip, inverter only	92%	
round trip, measured	82–83%	

The technology note's costs, the 2025 system capped at BNEF's turnkey price. Measured: the US battery fleet in 2019 (EIA) and PNNL's 2024 database, grid to grid.

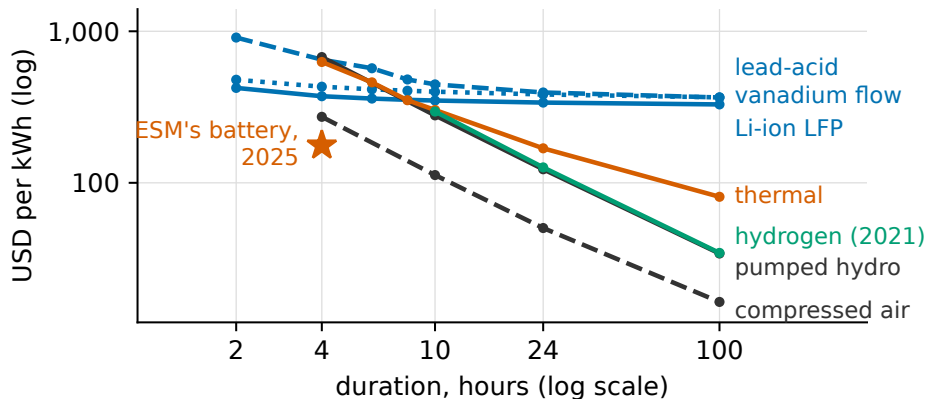
The two are bought separately, so they are sized separately. Part 2 asks which hours pay for each.

Every store is a price per MW and a price per MWh



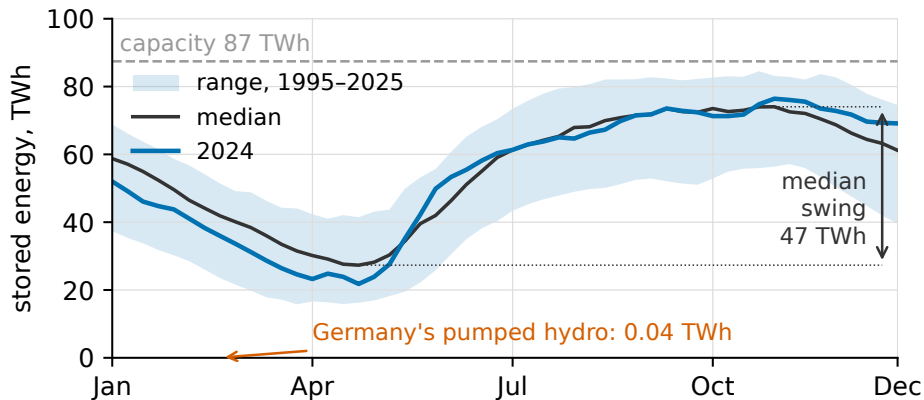
Reservoir hydro is off the chart: its tank is a lake, built long ago and filled by the weather. PNNL's storage database v2024, 100 MW projects, 2023 USD (hydrogen 2021).

Which store is cheapest depends on the duration



PNNL's storage database v2024, 100 MW, 2023 USD (hydrogen 2021). Star: ESM's 4-hour battery, 2025 European turnkey price.

Norway's reservoirs: a seasonal store at scale



Stored energy in Norway's reservoirs, week by week: NVE, reservoir statistics, the whole country, 1995-2025.

Hydro: cheap to own, and nearly built out

Cost-effective

- New reservoir hydro costs 52 EUR/MWh, against 39 for onshore wind. Almost all of it is the investment.
- Once built, it runs for 80 years, burns no fuel and costs 6.5 EUR/MWh to keep.
- The seasonal store comes with it: the tank is the lake, filled by rain and snow.

Further potential

- **Norway:** of 216 TWh a year that could be built, 64% is built and 23% protected. 23 TWh is left, 7.6 of it in upgrades.
- **Europe:** about half the technical potential is built, the highest share of any continent, best sites first.
- **What grows:** upgrades, and pumps between existing lakes. More MW, few new TWh.

Europe can add machines to the lakes it has. It cannot build another Norway.

Generation Technologies (at 40% utilisation); NVE (2020); JRC (2025).

Storage technologies: time scales and costs

	time scale	machine USD/kW	tank USD/kWh	round trip	in the course
lithium-ion battery	hours	209	327	83%	today, part 3
flow battery	hours	1,085	355	65%	
pumped hydro	hours to days	2,719	7	80%	today's paper
compressed air	hours to days	1,069	6	55%	
hydrogen in caverns	days to seasons	2,901	6	31%	lecture 7
reservoir hydro	seasons	2,430	a lake	–	lecture 5

PNNL's storage database v2024, 2023 USD (hydrogen 2021); reservoir hydro: *Generation Technologies*, a new plant.

Cheap machine, costly tank: hours. **Costly machine, cheap tank:** days and longer.

Workshop: Zerrahn et al. (2018)

Energy System Models, section 4: the storage model and its Lagrangian (§4.1), the arbitrage band, the table of duals.

Zerrahn et al. (2018), *On the economics of electrical storage for variable renewable energy sources*. Sections 1–5; §6 for the discussion.

Background: *Generation Technologies*, the batteries and *storage and prosumers* subsections. Optional: Butters et al. (2025), on whether arbitrage revenue pays for a battery.

Ambec and Crampes (2021), on flexible demand, moved to a block of its own later in the course.

The question

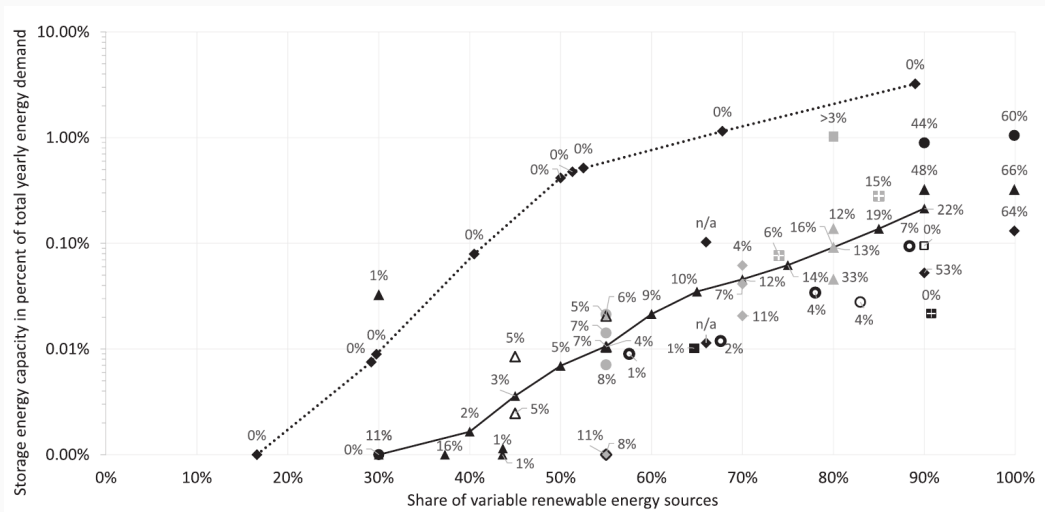
Sinn (2017): integrating wind and solar in Germany without waste needs a storage that takes every surplus MWh.

renewable share	50%	68%	89%
storage, GWh	2,100	5,800	16,300
share of annual demand	0.42%	1.15%	3.23%

Germany's pumped hydro holds just under 40 GWh.

Zerrahn et al. (2018): at 50%, a cost-minimising planner builds 35 GWh. **Which assumption produces a factor of sixty?**

The paper's Fig. 1: every study against Sinn



Zerrahn et al. (2018), Fig. 1, CC BY 4.0. Dotted: Sinn (2017). Labels: each study's curtailment.

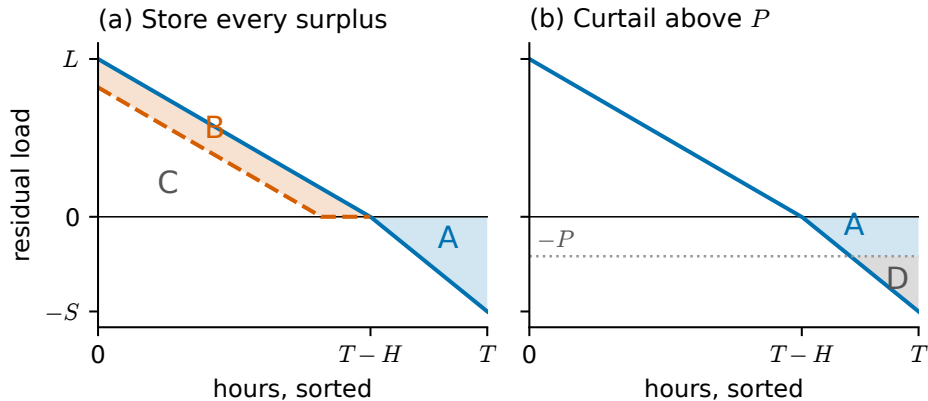
Part 1

Setup: a stylised residual load duration curve

- Residual load, load minus wind and solar, for every hour of the year, sorted from highest to lowest: the paper's device (§3.4).
- Below zero on the right, a surplus rising linearly to S MW over H hours. Storage efficiencies η_c and η_d .
- (a) Sinn's rule: store all of area A, release it as B; C is left to conventional plants.
- (b) A threshold P : curtail area D.

The curve is on the next slide and on the worksheet.

Setup: the curve



Why storage needs are two orders of magnitude below Sinn's

1. Sinn's claim, the two assumptions the authors alter, Fig. 1.
2. Sinn's rule on the curve: what passes through the tank; how the curve treats time.
3. The two curtailment strategies; $P = S(1 - \sqrt{q})$; why curtailment is efficient.
4. *Perspective*: Sinn's myopic storage against ESM's, and what foresight buys.

Stop point

The two assumptions; $\frac{1}{2}HS$ through the tank; what sets the tank; P for 5% and 20%.

The full questions, the hints and the stretch are on the worksheet.

Solutions 1

Solutions 1, Q1–Q2: the questions

- Q1. What does Sinn claim, and which two assumptions do the authors alter? What does Fig. 1 show?
- Q2. Sinn's heuristic on the curve: the areas, and how much energy passes through the tank. How does the curve treat time, and what does that mean for the tank?

Under Sinn's rule, what does the duration curve tell you about the size of the tank?

1. the tank is area A, the surplus energy
2. the tank is the peak surplus S times one hour
3. neither: the tank depends on the order of the hours, which the curve has thrown away

Solutions 1, Q1–Q2: the claim, and the curve

- **Q1. Two assumptions (p. 260):** Sinn looks only at the two corners, every surplus stored or none; and he minimises the tank, not the cost.
- **Q1. Fig. 1:** every other study curtails something, and finds storage an order of magnitude smaller.
- **Q2. Through the tank:** area A is $\frac{1}{2}HS$ a year, and $\eta_c\eta_d$ of it comes back as B.
- **Q2. Area A is not the tank:** the curve has no order, and “extended periods of renewable surpluses in contiguous hours determine the maximum energy capacity” (§4.2, p. 267).

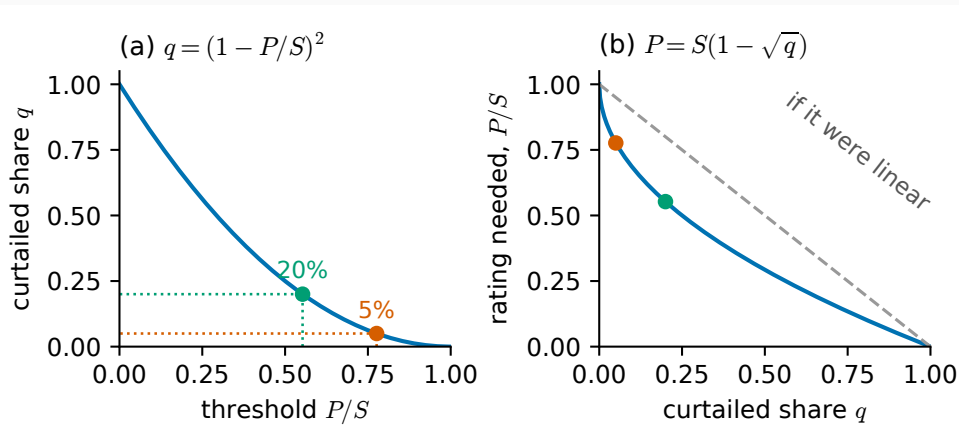
- Q3. The two curtailment strategies and what each targets; the tank at 50% renewables and 5% curtailment. Show that a threshold P curtails $q = (1 - P/S)^2$; P for 5% and 20%; the slope at $q = 0$; and why that makes some curtailment efficient.

Solutions 1, Q3: the square-root rule

- **Q3. Power-oriented** curtails above a MW threshold: 2.1 TWh $\rightarrow$ 0.3 TWh at 50%, 5% curtailed.
- **Q3. Energy-oriented** curtails when the tank is full: 2.1 TWh $\rightarrow$ **19 GWh**. It cuts the runs that set the tank.
- **Q3. The rule:** $P = S(1 - \sqrt{q})$: 0.78 S at 5%, 0.55 S at 20%. The slope is infinite at $q = 0$.
- **Q3. Efficient (p. 261):** the last MW of rating catches a sliver of energy; a few more turbines, curtailed now and then, are cheaper.

Board: the curtailed triangle D, similar to A with every side scaled by $1 - P/S$. The rule drawn is on the next slide.

Solutions 1, Q3: the square-root rule, drawn



Q4. *Perspective:* ESM's storage holds energy for the best hour ahead; Sinn's releases it at the first positive residual load. Into which hours does each release? What does foresight buy, and where is market reality (footnote 29)?

- **ESM's storage** releases where θ_h is beaten by the price: the far left of the curve, the scarcest hours.
- **Sinn's** releases at the first positive residual load: somewhere in the middle. B looks like a band on top only because footnote 16 re-sorts the hours.
- **Foresight buys** the two values of \$5.2: displacing the most costly fuel, and shaving the peak a plant would otherwise be built for.
- **Market reality** “likely lies between the myopic and the perfect foresight cases” (footnote 29).

Part 2

Setup: the planner of §5

A year of hourly demand d_h , served at least cost. The planner chooses every capacity and every hour's dispatch:

$$\min i^{se} N^{se} + i^{sp} N^{sp} + \sum_c i^c N^c + i^r N^r + \sum_{c,h} v^c G_h^c + \sum_h v^s (S_h^{in} + S_h^{out})$$

- Subject to: the hourly balance (2); plant capacity (3a); the weather's energy to load, storage or curtailment (3b); the storage (4a)–(4d); and at least a share α of demand from renewables (5).
- Storage has two capacities, N^{sp} in MW and N^{se} in MWh, each with its own annuity. Curtailment is free.
- The worksheet prints the Lagrangian and the first-order conditions. Your job is to say what they mean.

The cost-minimising model

1. The problem and its multipliers; the capacity conditions and σ ; why the tank stops far below Sinn's; *perspective*: ESM's storage model.
2. Fig. 10: D_1 , D_2 , B_1 – B_3 ; the two values of storage.
3. Germany's numbers at 50 and 90%; why duration and curtailment rise with the share.
4. Power-to-x: what a flexible new load does to storage.

Stop point

Which hours pay for pumps, basin and turbines, and what σ is. Germany: 35 GWh at 50%, no long-term storage, 35 $\rightarrow$ 4 GWh with power-to-x.

The full questions, the hints and the stretch are on the worksheet.

Solutions 2

- Q1. The problem and its multipliers. The four capacity conditions and the condition for G_h^c in words; why the planner builds turbines whose output is sometimes worth nothing. Why the tank stops far below Sinn's. *Perspective*: against ESM's storage model.

The annuity of one more MWh of basin is paid for by

1. every hour in which the basin holds some energy
2. the hours in which the basin is full
3. the hours in which it discharges at the highest prices

Solutions 2, Q1: which hours pay for what

a unit of	condition	is paid for by
pumps, N^{sp}	$i^{sp} = \sum_h (\bar{\phi}_h^c + \bar{\phi}_h^d)$	the hours at full power
basin, N^{se}	$i^{se} = \sum_h \bar{\psi}_h$	the hours the tank is full
turbines, N^r	$i^r = \sum_h \gamma_h \nu_h$	the hours its output is not curtailed
plant, N^c	$i^c = \sum_h \bar{\mu}_{c,h}$	the hours it runs flat out
conventional MWh	$\lambda_h = v^c + \sigma + \bar{\mu}_{c,h}$	σ is a tax on it

- **Q1. Curtailed hours:** $\nu_h = 0$, the spilled MWh was worth nothing; the turbine is paid by its other hours.
- **Q1. Perspective:** the operating conditions are ESM's arbitrage band; what is new is the investment layer.

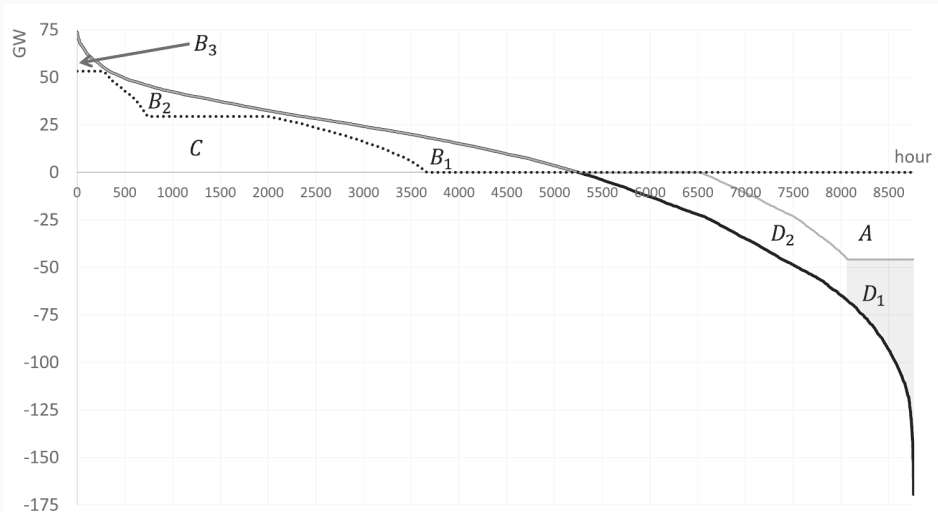
Solutions 2, Q1: why the tank stops far below Sinn's

1. The last MWh of basin is full only at the end of the **longest surplus runs**.
2. A skewed surplus has **few such runs**: a handful of full-tank hours a year, a handful of rents, far short of i^{se} .
3. So the planner stops building basin, and **curtails** the rest of the run, at no cost at the margin: those MWh have $\nu_h = 0$.

Sinn's tank is the one a planner builds if a MWh of basin is free, $i^{se} = 0$.

- Q2. In the paper's words, the areas D_1 , D_2 , B_1 – B_3 of Fig. 10; the two economic values of storage, and which areas carry each.
- Q3. Germany at 50% and 90%: energy, power, duration, and the conclusion on long-term storage. Why duration and curtailment rise.

Solutions 2, Q2: the paper's Fig. 10



- Q2. D_1 : curtailed above the 46 GW rating.
- Q2. D_2 : curtailed when the tank is full.
- Q2. B_1, B_2 : base and peak fuel displaced, the *arbitrage value*.
- Q2. B_3 : peak plant not built, the *capacity value*.

Solutions 2, Q3: Germany, and no long-term storage

renewable share	energy, GWh	power, GW	duration, h	curtailment
50%	35	6	6	5%
80%	~460			16%
90%	~1,080	58	19	~22%

The paper's §5.3; the 80 and 90% rows read off its Fig. 11.

- **Q3. The conclusion (p. 271):** “no need for a true long-term storage arises, that is, storing energy for weeks or months.”
- **Q3. Duration rises:** the MW is paid by the height of the spikes, the MWh by the length of the runs, and runs grow faster.
- **Q3. Curtailment rises:** the extra surplus comes in ever longer runs, and spilling it is cheaper than a tank to hold it.

Q4. Power-to-x (§5.4): what is added, and what happens to storage and curtailment at 50% and 90%? What does a shiftable load share with a storage, and what does it lack?

Solutions 2, Q4: a storage without a tank

- **Added:** 50 GW of perfectly flexible load, 2,000 full-load hours, 100 TWh a year, served by extra renewables.
- **At 50%:** storage 35 → **4 GWh**, curtailment 5% → under 1%. The new load eats the surplus.
- **At 90%:** storage unchanged. The load feeds on surplus that was being curtailed anyway, not on what the tank held.
- **The sentence:** a shiftable load consumes area A where it occurs. It is a storage without the tank, its cost and its losses, but it never gives the energy back.

Wrap-up

Closing: the planner on DK1 (part 3, optional)

The paper's model, equation for equation, on DK1's 2024 load and weather, with ESM's 2050 costs and its lithium-ion battery.

at 80% renewables	DK1	Germany (paper)
storage, share of annual demand	0.066%	0.092%
storage, hours of average load	5.8	
curtailment	14.9%	16%
renewables built without a target	68%	below 35%

Germany: the paper's §5.3 and Fig. 11. Part 3 of the worksheet works this, with two of §6's claims tested on DK1: one-way imports raise storage, and free charging changes nothing. The solutions work it in full.

What carries forward

1. **Flexibility is the problem, not generation cost.** Wind and solar win on levelised cost and lose value past about 40%, unless some other term of the balance can move.
2. **A storage is two capacities, paid by different hours.** Power by the hours at full power, energy by the hours the tank is full; their price ratio sets the duration.
3. **Curtailement is storage's partner.** A skewed surplus fills the last MWh of tank too rarely to pay for it, so storage needs are two orders of magnitude below Sinn's and never seasonal.

Lecture 5 takes two more of the four ways to balance: flexible demand in the heat sector, and trade over wires.

Before lecture 5 (Monday 5 October): *Energy System Models*, sections 5 and 6; Ruhnau et al. (2020) and Gonzales et al. (2023). Full solutions to today's worksheet, part 3 included, go up after class.

- Ambec, Stefan and Claude Crampes (2021). **“Real-Time Electricity Pricing to Balance Green Energy Intermittency”**. *Energy Economics* 94, p. 105074.
- Butters, R. Andrew, Jackson Dorsey, and Gautam Gowrisankaran (2025). **“Soaking up the Sun: Battery Investment, Renewable Energy, and Market Equilibrium”**. *Econometrica* 93.3, pp. 891–927.
- Gonzales, Luis E., Koichiro Ito, and Mar Reguant (2023). **“The Investment Effects of Market Integration: Evidence from Renewable Energy Expansion in Chile”**. *Econometrica* 91.5, pp. 1659–1693.
- Ruhnau, Oliver, Lion Hirth, and Aaron Praktiknjo (2020). **“Heating with Wind: Economics of Heat Pumps and Variable Renewables”**. *Energy Economics* 92, p. 104967.
- Zerrahn, Alexander, Wolf-Peter Schill, and Claudia Kemfert (2018). **“On the Economics of Electrical Storage for Variable Renewable Energy Sources”**. *European Economic Review* 108, pp. 259–279.